

1. Introduction

Proto-role approach:

Proto-agent and Proto-patient

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Sentence
Volition
Causation

(Dowty, 1991)

Thematic role list approaches:

Agent, Cause, Experiencer,
Theme, etc.

(Belletti & Rizzi, 1988; Fillmore, 1968;
Levin & Rappaport, 2005; Pesetsky,
1995)

In a previous eye-tracking reading experiment, we investigated the processing correlates of argument structures of *frighten*-type psych verbs:

- Proto-role approach predicts **competition**:

El marinero preocupó al vendedor.
The sailor worried the salesman.

↓
Sentence
Volition
Causation

- Thematic role list approach predicts **no competition**:

Agent Experiencer

We found:

agent-experiencer > experiencer-theme ≈ agent-theme

These results show evidence of competition for subjecthood between both arguments of *frighten*-type psych verbs

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Evidence in favor of proto-role approach

2. Experiment

Does the number of entailments modulate the competition between arguments of *frighten*-type psych verbs?

El marinero preocupó al vendedor.
The sailor worried the salesman.

↓
Sentence
Volition
Causation

agent - experiencer

El marinero golpeó al vendedor.
The sailor hit the salesman.

↓
Sentence
Volition
Causation

agent - theme

El viento preocupó al vendedor.
The wind worried the salesman.

↓
Causation

cause- experiencer

El viento golpeó al vendedor.
The wind hit the salesman.

↓
Causation

cause - theme

5. Discussion

Processing argument structures with competing arguments involves higher processing costs than processing argument structures without competing arguments.

Competition effect

When both arguments have proto-agent entailments, the difference in the number of entailments between both does not modulate processing cost.

This research provides evidence in favor of the proto-role approach.

El marinero preocupó al vendedor.

↓
Sentence
Volition
Causation

Proto-agent

El viento preocupó al vendedor.

↓
Causation

Proto-agent

No evidence of specific processing correlates for the experiencer role.

We provide evidence that there are specific processing correlates for proto-agent and proto-patient categories with no clear evidence for other role categories.

References

- Belletti, A., & Rizzi, L. (1988). Psych-Verbs and θ-Theory. *Natural Language & Linguistic Theory*, 6(3), 291–352.
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Fillmore, C. J. (1968). The Case for Case. In *Universals in Linguistic Theory* (pp. 1–88).
Levin, B., & Rappaport Hovav, M. (2005). *Argument realization* (1st published, repr). Cambridge University Press.
Pesetsky, D. M. (1995). *Zero Syntax: Experiencers and Cascades*. MIT Press.
Sánchez-López et al. (2023). Dowty was right: eye-tracking experiencers among agents and themes. *AMLAP*, San Sebastian, Spain.

3. Methods

Eye-tracking

49 Spanish native speakers (M=22 years old).

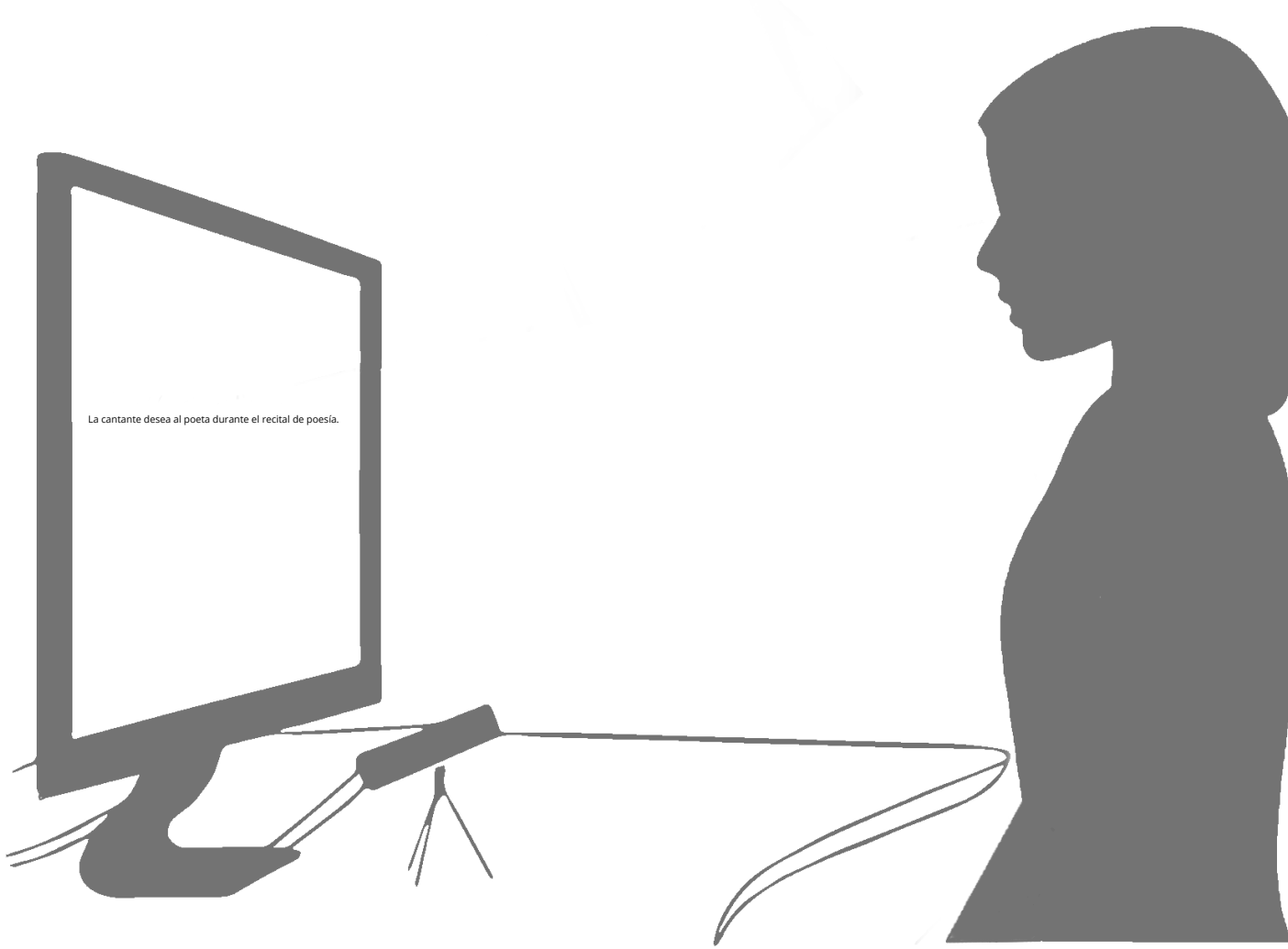
Reading task

20 verbs per condition controlled by length and frequency.

40 experimental items and 80 fillers.

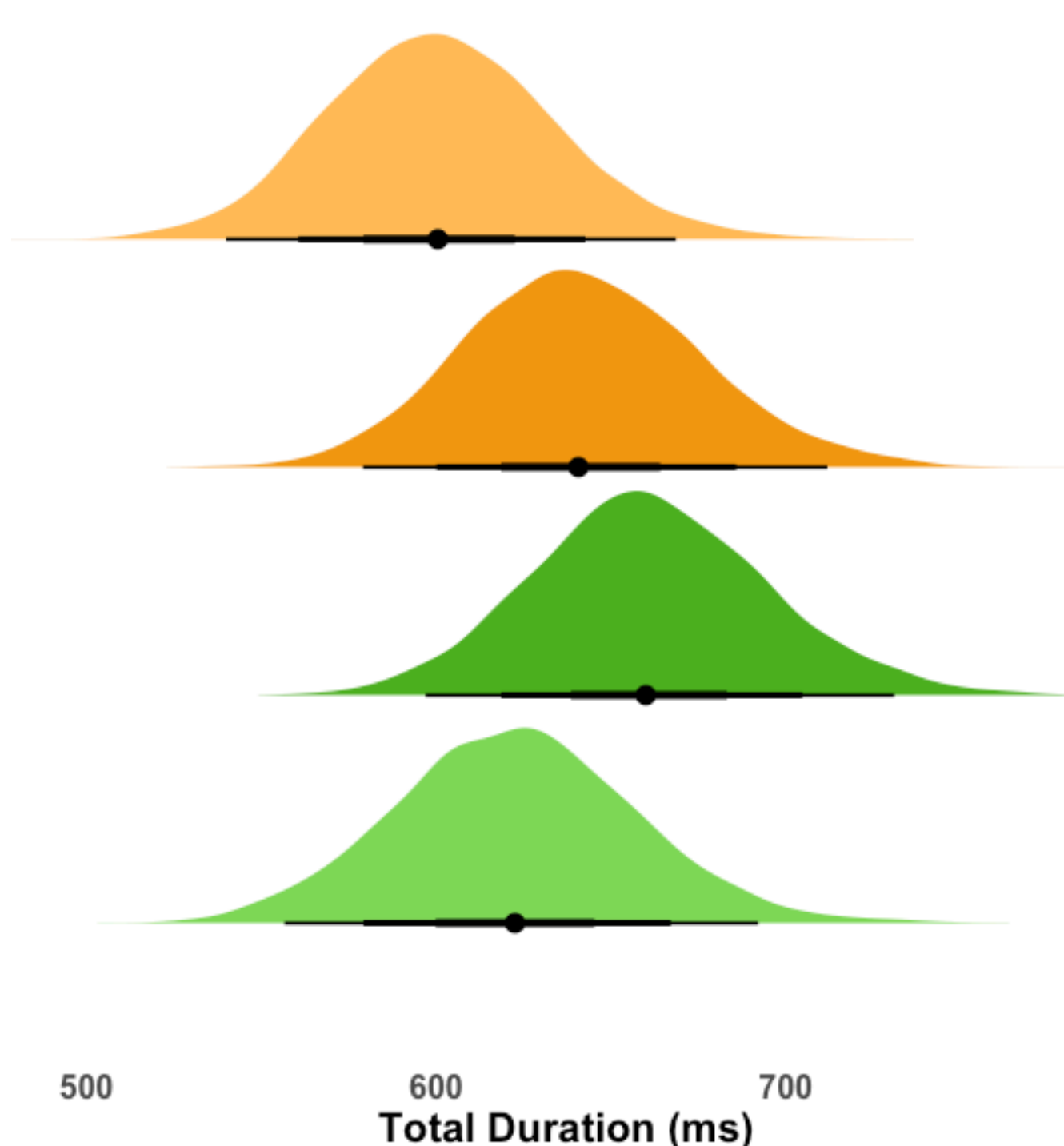
2x2 repeated measures design:

- **Subject Type** (agent vs. cause)
- **Object Type** (experiencer vs. theme)



4. Results

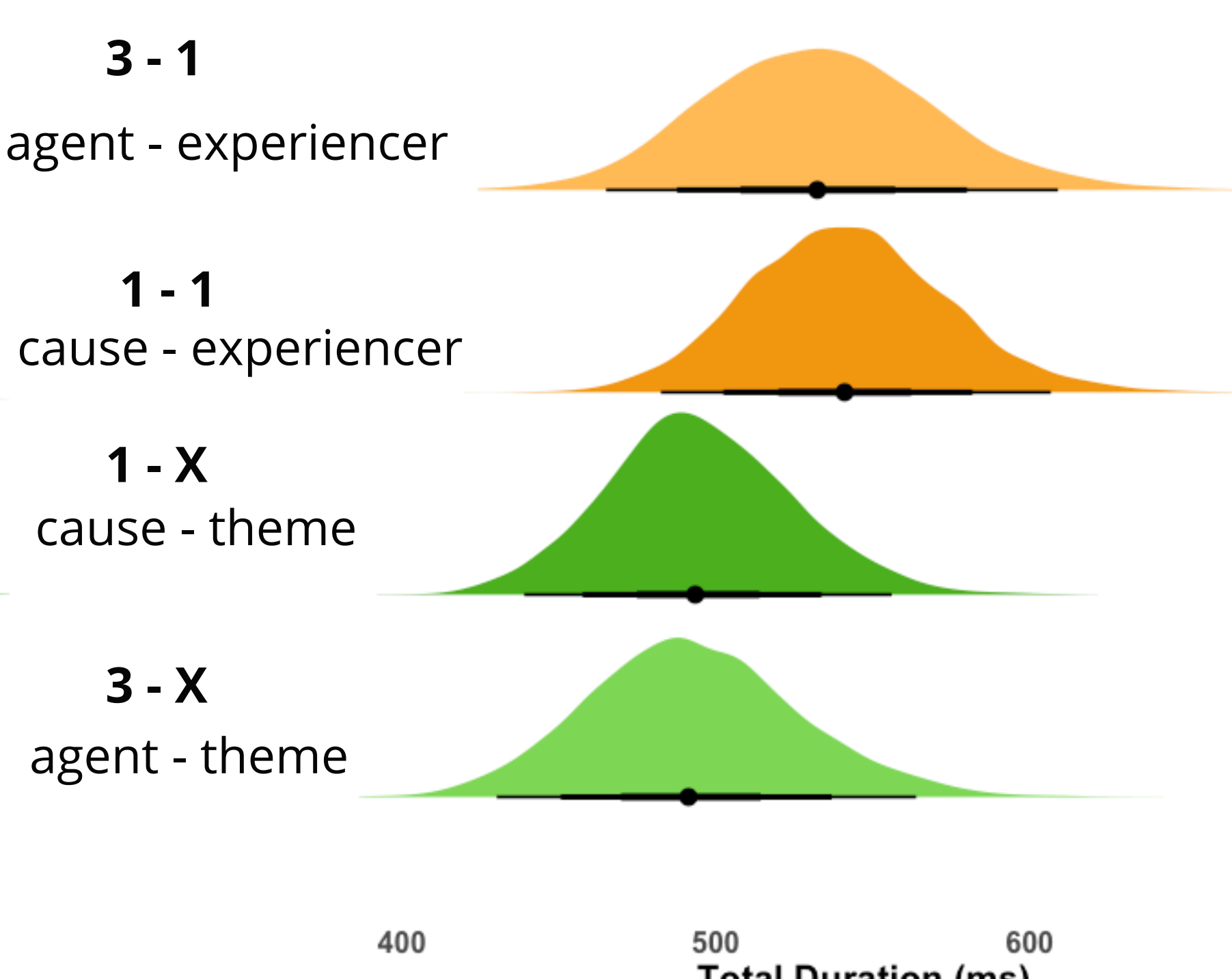
Subject Region



In the subject region, we found in intermediate/late measures:

- An animacy effect in the subject region.

Verb Region



In the verb region, we found in intermediate/late measures:

- **Agent/cause-experiencer** conditions involve a higher processing load than **agent/cause-theme**, independently of the subject type.
- No processing differences between **agent-experiencer** and **cause-experiencer**.

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